

SAMPLE REPORT — ILLUSTRATIVE EXAMPLE

PREPARED RESEARCH MEMORANDUM

Public Availability Assessment: Low-Rank Adaptation (LoRA) Fine-Tuning Technique

Cross-referencing verified, on-chain-attested AI model knowledge cutoffs against documented public disclosure milestones

PREPARED FOR	Meridian AI Labs, Inc. — Illustrative Example
PREPARED BY	Weight Registry (weightregistry.org)
RE	Public knowledge timeline support, trade secret / IP matter
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This is a sample work product illustrating the format and methodology of a Weight Registry commissioned research engagement. The client name above is fictional. The underlying historical facts, registered model data, and cryptographic hashes cited in this report are real and independently verifiable; the specific model responses shown are illustrative examples of the expected output pattern, not transcripts of a live inference session. This document does not constitute legal advice.

I. Executive Summary

This memorandum was prepared to support an assessment of when the Low-Rank Adaptation ("LoRA") fine-tuning technique became publicly available information within the AI/machine-learning community. Weight Registry cross-referenced the documented public disclosure history of LoRA against five independently verified, cryptographically hashed and timestamped open-source AI model checkpoints spanning the period in question, each selected for a knowledge cutoff date falling at a distinct point relative to LoRA's public emergence.

The analysis corroborates a public disclosure timeline beginning with LoRA's original arXiv preprint (June 17, 2021), continuing through its formal peer-reviewed presentation at ICLR 2022 (April 2022), and reaching broad mainstream tooling adoption with Hugging Face's PEFT library release (February 10, 2023). Models with verified knowledge cutoffs before the original preprint show no awareness of the technique; models cutoff after ICLR 2022 but before the PEFT release show awareness consistent with research-community-level disclosure; models cutoff after February 2023 treat the technique as an established, widely-adopted method.

II. Engagement Scope & Methodology

Weight Registry is a permissionless registry that cryptographically hashes open-source AI model weight files (SHA-256, SHA3-256, and BLAKE3) and timestamps the resulting commitment via a non-revocable attestation on Ethereum, using the Ethereum Attestation Service (EAS). Because each registered model's exact file contents and claimed knowledge cutoff are independently verifiable on-chain, a model's demonstrated awareness (or lack thereof) of a given topic can be used as a corroborating data point for when that information became part of the public record.

For this engagement, five models were selected from the registry on the basis of their verified knowledge cutoff dates, chosen to bracket each milestone in LoRA's documented public disclosure history. Each model was queried on its familiarity with the LoRA fine-tuning technique; responses were recorded alongside the model's on-chain attestation reference for independent verification.

III. Background: The Technique in Question

Low-Rank Adaptation (LoRA) is a parameter-efficient fine-tuning method for large language models. Rather than updating all of a pretrained model's weights during fine-tuning, LoRA freezes the original weights and injects small trainable low-rank matrices into each layer of the Transformer architecture, substantially reducing the compute and storage required to adapt a large model to a new task. Questions of when such a technique

became publicly known information are frequently material in trade secret misappropriation defenses, patent obviousness challenges, and independent-invention claims, where a party's awareness (or a technique's general availability) as of a specific date is at issue.

IV. Documented Public Disclosure Timeline

The following milestones in LoRA's public disclosure history are independently sourced and verifiable; none are estimated.

DATE	EVENT	SOURCE
June 17, 2021	Original paper, "LoRA: Low-Rank Adaptation of Large Language Models" (Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang, Chen — Microsoft), submitted to arXiv	arXiv:2106.09685
October 16, 2021	Revised version posted to arXiv (same identifier)	arXiv:2106.09685
April 25–29, 2022	Accepted and presented as a peer-reviewed conference paper at ICLR 2022 (virtual)	OpenReview id=nZeVKeeFYf9
February 10, 2023	Hugging Face announces the PEFT (Parameter-Efficient Fine-Tuning) library with first-class LoRA support — the major broad-adoption milestone making LoRA fine-tuning accessible to the mainstream open-source ML community	huggingface.co/blog/peft

V. Model Cross-Reference: Verified Knowledge by Cutoff Date

The responses shown below are illustrative examples demonstrating the expected pattern of model awareness relative to each verified knowledge cutoff date, consistent with the documented disclosure timeline in Section IV. They are not transcripts of a live inference session. In a commissioned engagement, Weight Registry runs the query against the actual registered checkpoint and delivers the model's genuine output.

GPT-J-6B

ON-CHAIN · ELEUTHERAI

Verified knowledge cutoff: **2021-06-04** — approximately two weeks before LoRA's original preprint

QUERY

"Are you familiar with the LoRA (Low-Rank Adaptation) fine-tuning technique for large language models?"

ILLUSTRATIVE RESPONSE

"I'm not familiar with a technique by that name. Could you describe what you mean?"

Combined SHA-256: 0xf051f5cc39418523098d3eb980bd2dc8b2c36524eb01f953089cdb09bb6c50a1 · Explorer: weightregistry.org/explorer/gpt-j-6b/

BLOOM-176b

PENDING ATTESTATION · BIGSCIENCE

Verified knowledge cutoff: **2022-07-11** — after ICLR 2022 acceptance, before the PEFT library release

QUERY

"Are you familiar with the LoRA (Low-Rank Adaptation) fine-tuning technique for large language models?"

ILLUSTRATIVE RESPONSE

"Yes — LoRA is a method proposed by Microsoft researchers that freezes the pretrained model weights and injects trainable low-rank decomposition matrices into the Transformer layers, reducing the number of trainable parameters needed for fine-tuning. It was presented at ICLR 2022."

Weights SHA-256 (Merkle root, 72 shards):
9a4612545e7d2636d8e12ac036005913f89518fab147675b867d0c07f1a99c5e · Explorer: weightregistry.org/explorer/bloom-176b/

Vicuna-7b-v1.5

ON-CHAIN · LMSYS

Verified knowledge cutoff: **2023-08-01** — after the PEFT library release

QUERY

"Are you familiar with the LoRA (Low-Rank Adaptation) fine-tuning technique for large language models?"

ILLUSTRATIVE RESPONSE

"Yes, LoRA is a widely-used parameter-efficient fine-tuning method. It's well supported in the open-source ecosystem — for example, Hugging Face's PEFT library makes it straightforward to apply LoRA to most pretrained models without a custom implementation."

Combined SHA-256: 0x6dca84c834b561e6f0c6903606ce5aa7c0d92050ceca127a67c6f29040673424 · Explorer: weightregistry.org/explorer/vicuna-7b-v1-5/

Mistral-7B-v0.1

ON-CHAIN · MISTRAL AI

Verified knowledge cutoff: **2023-09-27** — after the PEFT library release

QUERY

"Are you familiar with the LoRA (Low-Rank Adaptation) fine-tuning technique for large language models?"

ILLUSTRATIVE RESPONSE

"Yes. LoRA is a standard approach for efficient fine-tuning at this point, commonly used to adapt base models like this one for downstream tasks without the cost of full fine-tuning."

Combined SHA-256: 0x946c3055d0bcedbc9ac13f9d5a5a3dc5b1a27935f685d4f380557b7aa9e69c9f · Explorer: weightregistry.org/explorer/mistral-7b-v0-1/

Qwen2.5-7B

ON-CHAIN · ALIBABA

Verified knowledge cutoff: **2024-09-18** — well after the PEFT library release

QUERY

"Are you familiar with the LoRA (Low-Rank Adaptation) fine-tuning technique for large language models?"

ILLUSTRATIVE RESPONSE

"Yes — LoRA has been a baseline parameter-efficient fine-tuning method for several years now, and a number of variants and extensions have since been proposed building on the original approach."

Combined SHA-256: 0xd069f4287f844e5e0e2b50bdf8ca79c9b33ad2bc30ad86124ab064534dbd2d6f · Explorer: weightregistry.org/explorer/qwen2-5-7b/

VI. Synthesis & Findings

The pattern of verified model awareness aligns consistently with the documented disclosure timeline in Section IV:

1. **Before June 17, 2021:** No awareness demonstrated (GPT-J-6B, cutoff 2021-06-04).
2. **After ICLR 2022, before PEFT (April 2022 – February 2023):** Awareness demonstrated at a research/technical level, consistent with disclosure limited primarily to the academic and early-adopter AI research community (BLOOM-176b, cutoff 2022-07-11).
3. **After PEFT release (February 10, 2023 onward):** Awareness demonstrated as an established, mainstream, tooling-supported technique (Vicuna-7b-v1.5, Mistral-7B-v0.1, Qwen2.5-7B).

FINDING

This cross-reference corroborates that LoRA fine-tuning was public, documented information in the AI/ML research community no later than its April 2022 ICLR presentation, and had reached broad mainstream public availability no later than the February 10, 2023 release of Hugging Face's PEFT library.

VII. Limitations & Scope of This Analysis

- A registered model's demonstrated awareness of a topic is a corroborating data point, not direct proof of public availability on a specific date; it reflects what the model's training data contained as of its claimed cutoff.

- A model's claimed knowledge cutoff is a self-reported value, cryptographically bound to that model's on-chain attestation record. Weight Registry independently verifies the hash of the weights themselves; it does not independently verify the accuracy of a publisher's claimed cutoff date against the model's actual training data.
- This report illustrates Weight Registry's methodology and is not, and should not be construed as, legal advice or a legal conclusion. Parties relying on this type of analysis in an actual legal matter should do so in consultation with qualified counsel.

Appendix A Independent Verification

Every hash and attestation cited in this report can be independently verified without any involvement from Weight Registry. For any model above: (1) download its weights, config, and tokenizer files from the source repository listed on its Explorer page; (2) compute SHA-256, SHA3-256, and BLAKE3 digests locally; (3) compare against the values published on-chain via the Ethereum Attestation Service, independently of Weight Registry's own records. Step-by-step instructions are available at weightregistry.org/news/how-to-verify-ai-model-weights/.

Weight Registry — weightregistry.org — contact@weightregistry.org

This is a sample document prepared for illustrative purposes to demonstrate Weight Registry's commissioned research report format.